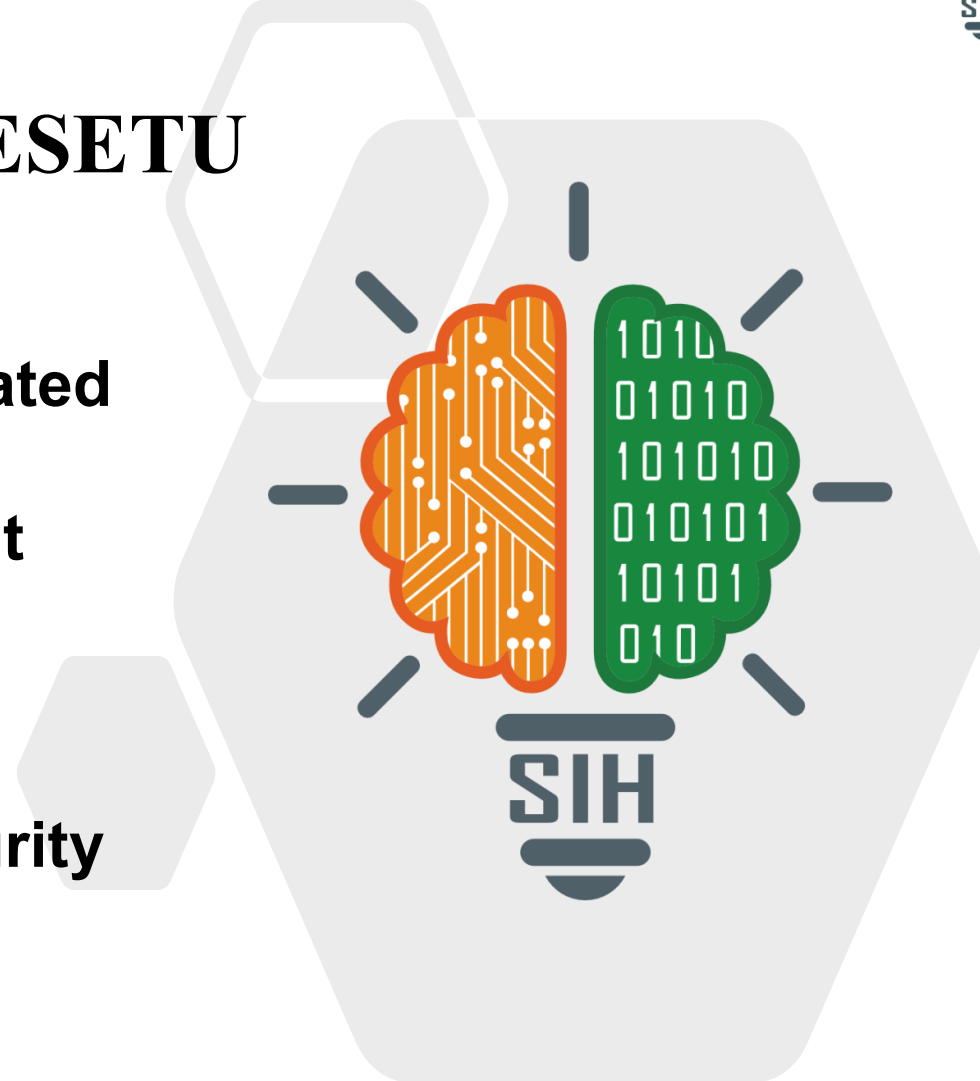


SMART INDIA HACKATHON 2026



TRACESETU

- **Problem Statement ID – PS26182**
- **Problem Statement Title – Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs**
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID – 170835**
- **Team Name – ANANTHA**



Find the first receiving crypto service

THE PROBLEM

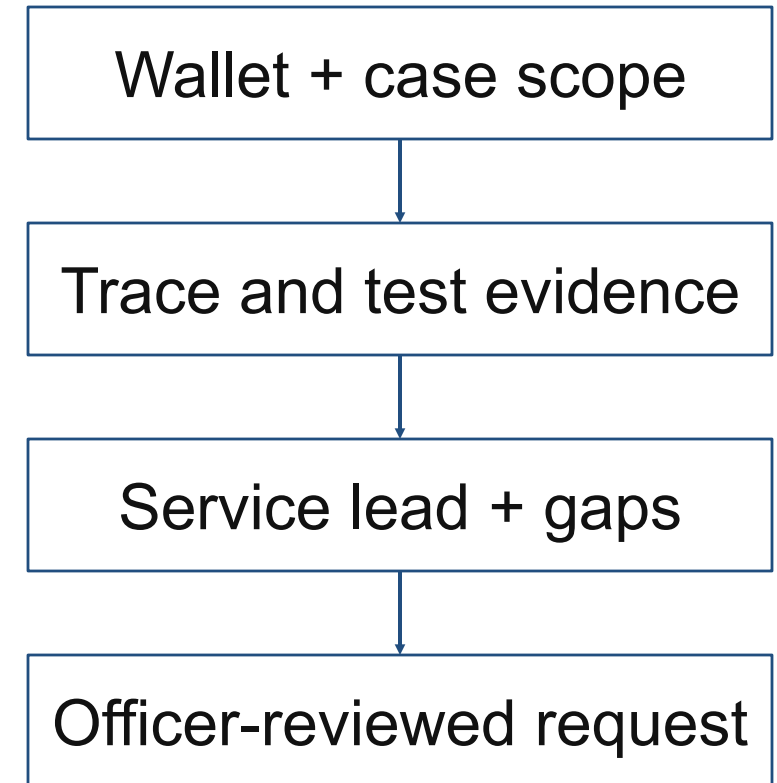
Transfers reveal addresses, but rarely the company behind them. The first deposit wallet may have no label.

OUR SOLUTION

Trace to the first supported exchange or custodial service on each path.

OUR DIFFERENCE

Test earlier deposit candidates. Show the evidence gap and the next check.



VASP = exchange or custodial service

1 READ SOURCES

Transfers, receipts
and sourced
service
wallets

2 PATTERNS [1, 2]

Token collection
plus service-funded
transaction fees

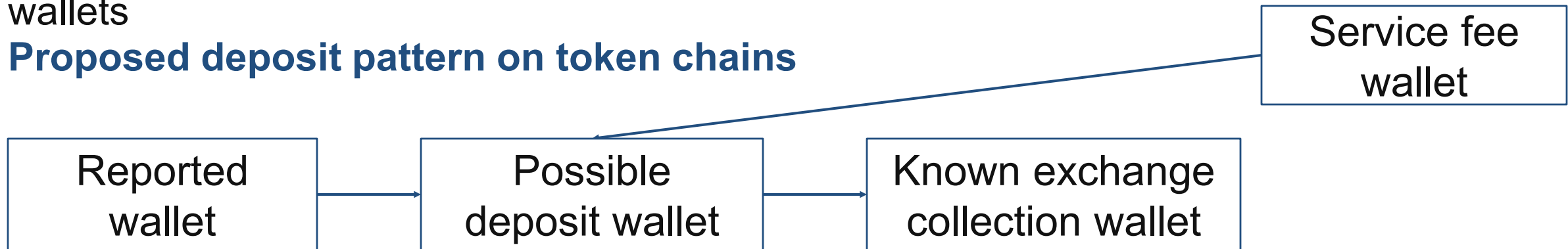
3 TRACE [3]

Fewest steps in
time order. Stop at
first custodian.

4 REVIEW

Inspect paths,
sources and gaps
before acting.

Proposed deposit pattern on token chains



Repeated collection + service fee funding suggest a role, not ownership.

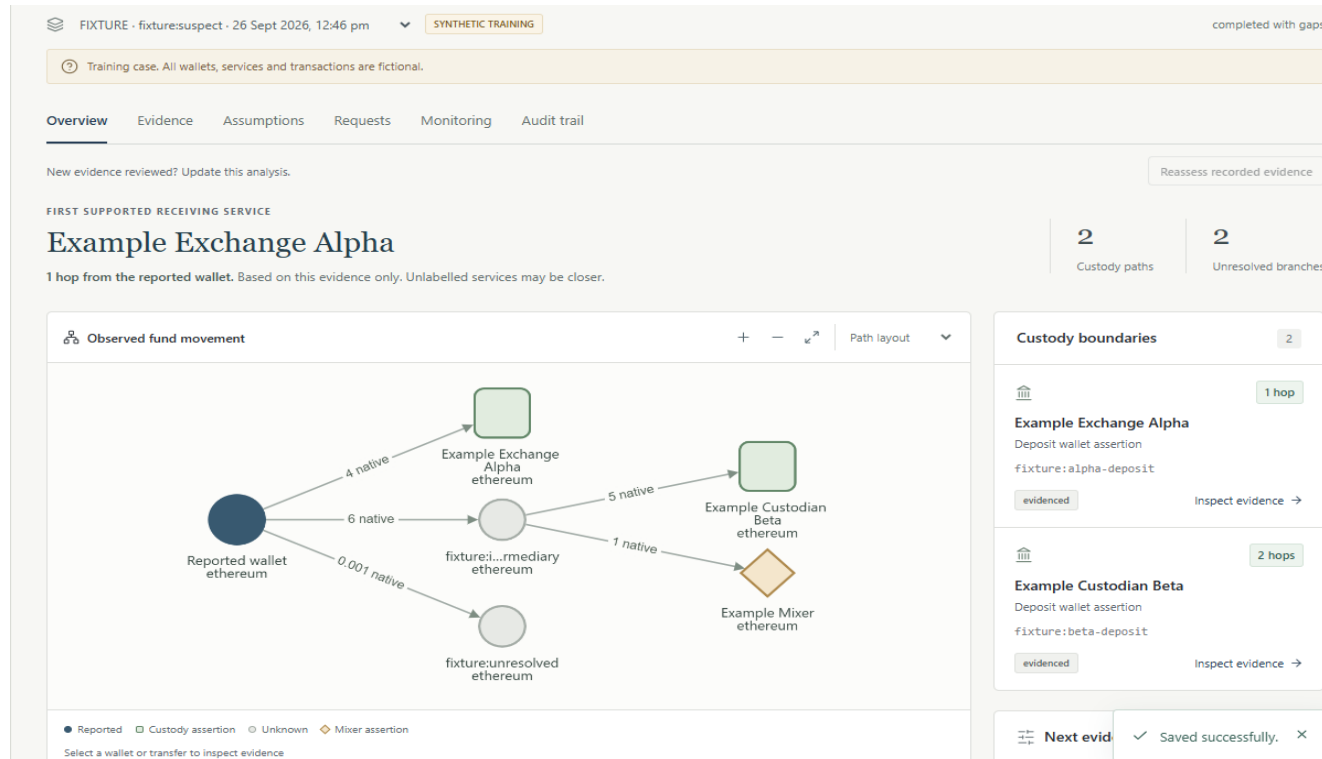
Python / FastAPI

React / TypeScript

SQLite

Cytoscape graph view

WORKING PROTOTYPE



Actual interface. Synthetic training case.

BTC / ETH / BNB / Polygon / Tron / Solana adapters. Public/free APIs. SAHYOG not connected.

BUILT AND TESTED

Case tracing, source challenge, query budgets and signed evidence export.

RESEARCH CHECK

Separate detector: 64 synthetic tests pass. MVP integration and independent live validation pending.

RISK AND CONTROL

Customers can imitate deposits. Retain a hypothesis until independently confirmed.

A better-supported decision about whom to approach

INVESTIGATOR See the receiving-service lead, its path and missing evidence.

REVIEWER Remove a disputed source. Recompute without rewriting history.

NEXT ACTION Prioritise useful queries. Prepare a request for a verified recipient.

HOW WE WILL MEASURE VALUE [5]

Deposit-role precision, false links, answer coverage, API calls and officer time.
Compare with label-only tracing using independent service-confirmed records.

1 Deposit-address patterns

[Victor, FC 2020](#)

2 Service fee funding + collection

[Brechlin et al., 2025](#)

3 Time-respecting graph paths

[Wu et al., PVLDB 2014](#)

4 Bitcoin clustering (planned)

[Kappos et al., USENIX 2022](#)

5 Independent attribution evaluation

[Lubbertsen et al., USENIX 2025](#)

[PUBLIC SOURCE CODE + RESEARCH](#)

github.com/notvinayaks/TraceSetu

[WATCH THE PROTOTYPE](#)

youtube.com/watch?v=V_qB6dL934w

Published methods guide the design. Synthetic tests do not establish real-world accuracy.